

Cell & Molecular Biology
Lab Activity:

Characterization of a Plastid Genome

Data Source and Genome Selection

Organism	<i>Curcuma longa</i> (turmeric)
Family	Zingiberaceae
NCBI Accession/Version	NC_042886.1
Database Source	NCBI Reference Sequence (RefSeq)
Genome length	159,550 bp
Topology	Circular
Associated reference/publication	Windsor et al. (2019), <i>Microbiology Resource Announcements</i>

Files to obtain: complete FASTA sequence and the annotated GenBank record for NC_042886.1. The annotated GenBank file was used to characterize gene features, while the complete sequence was used for sequence statistics.

Required Plastid Genome Characterization

Characteristic	Result
Chosen genus and species	<i>Curcuma longa</i>
Family	Zingiberaceae
NCBI accession/version	NC_042886.1
Complete genome size	159,550 bp
GC content	36.31%
Genome topology	Circular
LSC size	87,058 bp
SSC size	18,542 bp
IR size	26,975 bp each
Total annotated gene features	133
Protein-coding genes (CDS)	86
tRNA genes	38

rRNA feature records	8
Introns	Present; examples include <i>rps16</i> , <i>atpF</i> , <i>rpoC1</i> , <i>ycf3</i> , <i>clpP</i> , <i>petB</i> , <i>petD</i> , <i>ndhA</i> , <i>ndhB</i> , <i>rpl2</i> , <i>rps12</i> , <i>trnK-UUU</i> , <i>trnL-UAA</i>
Pseudogenes	One <i>ycf1</i> pseudogene annotation
Gene duplications	Present, mainly involving genes in the inverted-repeat regions
Sequence records	1

Gene Groups to Identify

Gene group	Genes found
Photosystem I (<i>psa</i>)	<i>psaA</i> , <i>psaB</i> , <i>psaC</i> , <i>psaI</i> , <i>psaJ</i>
Photosystem II (<i>psb</i>)	<i>psbA</i> , <i>psbB</i> , <i>psbC</i> , <i>psbD</i> , <i>psbE</i> , <i>psbF</i> , <i>psbH</i> , <i>psbI</i> , <i>psbJ</i> , <i>psbK</i> , <i>psbL</i> , <i>psbM</i> , <i>psbN</i> , <i>psbT</i> , <i>psbZ</i>
ATP synthase (<i>atp</i>)	<i>atpA</i> , <i>atpB</i> , <i>atpE</i> , <i>atpF</i> , <i>atpH</i> , <i>atpI</i>
Cytochrome b6f (<i>pet</i>)	<i>petA</i> , <i>petB</i> , <i>petD</i> , <i>petG</i> , <i>petL</i> , <i>petN</i>
Carbon fixation	<i>rbcL</i>
RNA polymerase (<i>rpo</i>)	<i>rpoA</i> , <i>rpoB</i> , <i>rpoC1</i> , <i>rpoC2</i>
Ribosomal proteins (<i>rpl</i>)	<i>rpl2</i> , <i>rpl14</i> , <i>rpl16</i> , <i>rpl20</i> , <i>rpl22</i> , <i>rpl23</i> , <i>rpl32</i> , <i>rpl33</i> , <i>rpl36</i>
Ribosomal proteins (<i>rps</i>)	<i>rps2</i> , <i>rps3</i> , <i>rps4</i> , <i>rps7</i> , <i>rps8</i> , <i>rps11</i> , <i>rps12</i> , <i>rps14</i> , <i>rps15</i> , <i>rps16</i> , <i>rps18</i> , <i>rps19</i>
rRNA (<i>rrn</i>)	<i>rrn4.5</i> , <i>rrn5</i> , <i>rrn16</i> , <i>rrn23</i>
tRNA (<i>trn</i>)	Multiple <i>trn</i> genes; 38 tRNA features are annotated
Other conserved genes	<i>matK</i> , <i>clpP</i> , <i>accD</i> , <i>cemA</i> , <i>ycf1</i> , <i>ycf2</i> , <i>ycf3</i> , <i>ycf4</i> , <i>infA</i> , <i>ccsA</i>

Genome Retrieval and Galaxy Analysis

The complete chloroplast record NC_042886.1 was retrieved from the NCBI Nucleotide/RefSeq database. The FASTA sequence and annotated GenBank file were retained for analysis. The FASTA sequence was uploaded to Galaxy using the history name **Plastid_Curcuma_Olaguir**. The FASTA Statistics tool (gfastats) or an equivalent sequence-statistics tool was used to characterize the sequence.

Sequence-statistics check for the exact NC_042886.1 sequence: 1 sequence record; 159,550 bp; GC content 36.31%; N50 159,550 bp; L50 1; N = 0. These values correspond to the complete single-record sequence and should be reproduced when the exact NC_042886.1 FASTA is used in Galaxy.

Important: an earlier Galaxy run showing 168,985 bp and 36.82% GC corresponds to a different plastid sequence and should not be used as the final *Curcuma* result. The final Galaxy screenshot should therefore be taken after uploading the exact NC_042886.1 FASTA.

Questions for the Student Report

1. Give the full scientific name, family, NCBI accession/version, database source, and complete plastid-genome size.

- The selected organism is *Curcuma longa*, commonly known as turmeric, a member of the family Zingiberaceae. Its complete chloroplast genome is available in the NCBI Reference Sequence (RefSeq) database under accession NC_042886.1. The complete plastid genome is 159,550 bp long and is circular.

2. What evidence shows that the sequence is a complete plastid/chloroplast genome rather than a barcode marker, genome fragment, or nuclear sequence?

- The NCBI record identifies the sequence as the complete chloroplast genome of *Curcuma longa* and provides the RefSeq accession NC_042886.1. The record is a single circular plastid sequence of 159,550 bp with extensive chloroplast gene annotation, including photosystem, ATP synthase, cytochrome b6f, ribosomal, tRNA, rRNA, and other conserved plastid genes. These features support that it is a complete plastome rather than a short barcode or nuclear fragment.

3. Describe the overall organization of the plastid genome. Does it contain the common LSC–IR–SSC–IR arrangement? Give the sizes of these regions when available.

- Yes. The *C. longa* plastome follows the common quadripartite organization: Large Single-Copy (LSC) – Inverted Repeat (IR) – Small Single-Copy (SSC) – IR. The LSC is 87,058 bp, the SSC is 18,542 bp, and each IR is approximately 26,975 bp. Together these regions account for the complete 159,550-bp plastome.

4. Summarize the annotated gene content: total genes, protein-coding genes, tRNA genes, rRNA genes, and pseudogenes. Explain why genes located in the inverted-repeat regions may appear in two copies.

- The current NC_042886.1 GenBank annotation contains 133 *gene* feature records, including 86 CDS features, 38 tRNA features, and 8 rRNA feature records. One *ycf1* feature is annotated as a pseudogene. The associated publication reports 125 genes under its counting convention; the difference reflects how duplicated IR genes and annotation records are counted. Genes within the IR regions can occur in two copies because the two IR regions are homologous, so features located in one IR may be represented again in the second IR.

5. Choose at least eight protein-coding plastid genes from different functional groups. List each gene and briefly explain its biological function.

- **psaA** – photosystem I reaction-center protein involved in light-driven electron transfer.
- **psbA** – photosystem II D1 protein, essential for the photosynthetic reaction center.
- **atpA** – ATP synthase alpha subunit involved in ATP production.
- **petA** – cytochrome f component of the cytochrome b6f complex involved in electron transport.
- **rbcL** – large subunit of Rubisco involved in carbon fixation.
- **rpoB** – plastid RNA polymerase subunit involved in transcription.
- **rps12** – ribosomal protein S12 involved in translation.

- **ndhA** – subunit of the chloroplast NADH dehydrogenase-like complex involved in photosynthetic electron transport.
- **matK** – maturase involved in RNA intron processing/splicing.

6. Identify important RNA and RNA-processing features. Include the rRNA genes, examples of tRNA genes, and at least two genes with introns if present in your genome.

- The annotation contains four major rRNA types: *rrn4.5*, *rrn5*, *rrn16*, and *rrn23*. Examples of tRNA genes include *trnK-UUU*, *trnQ-UUG*, *trnS-GCU*, *trnG-GCC*, *trnR-UCU*, and *trnH-GUG*. Intron-containing genes include *rps16*, *atpF*, *rpoC1*, *ycf3*, *clpP*, *petB*, *petD*, *ndhA*, *ndhB*, *rpl2*, *rps12*, *trnK-UUU*, *trnL-UAA*, *trnA-UGC*, and *trnV-UAC*. The presence of introns indicates that RNA processing and splicing are important parts of plastid gene expression.

7. Describe any pseudogenes, gene losses, duplications, rearrangements, or other unusual features reported for your plastid genome. If none are reported, state this clearly.

- The current GenBank annotation includes a *ycf1* pseudogene annotation. Gene duplications are prominent in the IR regions; examples include *rps12*, *rps19*, *rpl2*, *rpl23*, *ycf2*, *ndhB*, and *rps7* among the CDS features, as well as several duplicated tRNAs. The sources used for this report do not establish a major genome rearrangement or specific gene-loss event for this accession, so no such event is claimed here.

8. What is the GC content of your plastid genome? Based on the sequence statistics and annotation, describe two other notable sequence or structural observations.

- The overall GC content is 36.31%. Two notable observations are the typical LSC–IR–SSC–IR quadripartite structure and the high level of gene duplication associated with the IR regions. The exact sequence contains one complete circular record with no ambiguous N bases.

9. Compare plastid and mitochondrial genomes. Give at least five similarities and five differences, considering location, biological role, inheritance, genome organization, gene content, copy number, and evolutionary behavior.

- **Similarities:**

1. both are organellar genomes;
2. both originated through endosymbiosis;
3. both retain their own genetic material;
4. both are present in multiple copies in cells/organelles; and
5. both are useful in evolutionary and phylogenetic studies.

- **Differences:**

1. plastid genomes occur in chloroplasts/plastids, whereas mitochondrial genomes occur in mitochondria;
2. plastids contain genes strongly associated with photosynthesis and plastid gene expression, whereas mitochondria contain genes mainly associated with respiration and oxidative phosphorylation;
3. plastomes are commonly compact and structurally conserved with an LSC–IR–SSC–IR organization, whereas plant mitochondrial genomes are often much more structurally variable;
4. plastid gene content includes photosystem, *rbcL*, and many plastid ribosomal genes, while mitochondrial gene content differs substantially; and

5. plastid and mitochondrial genomes have different patterns of recombination, repeats, and structural evolution.

10. Explain the practical value of plastid genomes in research. List advantages compared with the nuclear genome and important limitations. Give one research question for which plastid data would be useful and one for which nuclear genomic data would be more appropriate.

- Plastid genomes are valuable for species identification, DNA barcoding, phylogenetic analysis, population studies, biodiversity assessment, and conservation. They are relatively small, compact, and often conserved, making them practical for comparative analyses. However, plastid data represent only the plastid lineage and may reflect uniparental inheritance, hybridization, or introgression rather than the full species history. A plastid-based question could be: *How is Curcuma longa related to other Curcuma species based on complete chloroplast genomes?* A nuclear-genome question would be: *Which nuclear genetic variants contribute to a complex trait or distinguish populations across the entire genome?*

Plastid vs Mitochondrial Genome Comparison

Feature	Plastid genome	Mitochondrial genome
Cellular location	Chloroplasts and other plastids	Mitochondria
Main biological functions	Photosynthesis, carbon fixation, and synthesis of some plastid proteins	Cellular respiration, oxidative phosphorylation, and mitochondrial metabolism
Typical genome organization	Usually circular in angiosperms, commonly with LSC–IR–SSC–IR quadripartite structure	Highly variable in plants; may be circular, linear, branched, or represented by multiple DNA molecules
Relative genome size	Usually smaller and more conserved; <i>C. longa</i> is 159,550 bp	Usually much larger and highly variable among plants
Gene content	Photosynthesis, transcription, translation, rRNAs, tRNAs, and other plastid functions	Respiration, oxidative phosphorylation, translation, and other mitochondrial functions
Copy number	Usually present in multiple copies per cell/organelle	Multiple copies per cell/organelle; copy number varies
Inheritance	Usually uniparental; commonly maternal in angiosperms, with exceptions	Usually uniparental; commonly maternal in angiosperms, with exceptions
Recombination / structural change	Generally more conserved, although structural rearrangements can occur	Generally much more structurally variable, with repeats and recombination contributing to rearrangements
Mutation / substitution pattern	Often relatively conserved and useful for phylogenetic studies	Protein-coding sequences can be conserved while genome structure changes substantially
Common research applications	DNA barcoding, species identification, phylogeny, plant evolution, biodiversity, and conservation	Plant evolution, cytoplasmic inheritance, mitochondrial function, fertility, and mitochondrial genome evolution

Reproducibility and Data Sources

Reproducibility steps

1. Open the NCBI Nucleotide/RefSeq record for NC_042886.1.
2. Download the complete FASTA sequence and annotated GenBank file.
3. Upload the FASTA file to Galaxy and create/use the history **Plastid_Curcuma_Olaguir**.
4. Run FASTA Statistics (gfastats) or an equivalent sequence-statistics tool.

5. Retain the output and screenshot showing the sequence length and GC statistics.
6. Use the annotated GenBank record to document gene groups, introns, pseudogene annotations, and duplicated genes.
7. Record all accession numbers, tool names, and file versions so another student can reproduce the analysis.

Data sources

- NCBI Reference Sequence: NC_042886.1 - *Curcuma longa* chloroplast, complete genome.
- Windsor, A. M., Ott, B. M., Zhang, N., Wen, J., Hsu, E., & Handy, S. M. (2019). Full Chloroplast Genome Sequence of the Economically Important Dietary Supplement and Spice *Curcuma longa*. *Microbiology Resource Announcements*, 8(32), e00576-19.
- Galaxy Project - used for sequence-statistics analysis and workflow documentation.